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Taking Science on Faith

A discourse in response to the essay by

Paul C. Davies

WITH RESPONSES FROM

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and a reply from Paul Davies

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In November 2007 the physicist and cosmologist Paul Davies published a New York Times op-ed arguing that science, like religion, rests on an act of faith: the belief that the universe is governed by dependable, universal laws. The essay drew sharp responses from physicists, biologists, philosophers, and science writers. Their replies, collected here from the Edge.org Reality Club, are followed by Davies's reply to his critics.

Jerry Coyne

Response to Paul Davies

With Paul Davies' article, the New York Times continues its tradition of soft-headed op-ed pieces that criticize science and, either implicitly or explicitly, bolster religion. I have just two comments:

1. Contrary to Davies' assertion, science is not based on "faith" that physical laws will apply forever, or in different places in the Universe. This is an observation—an observation that has not been contradicted by any other data. Davies is completely off base when claiming that "to be a scientist, you had to have faith that the universe is governed by dependable, immutable, absolute, universal, mathematical laws of an unspecified origin. You've got to believe that these laws won't fail, that we won't wake up tomorrow to find heat flowing from cold to hot, or the speed of light changing by the hour. " This is not a matter of faith. It's a matter of experience. In contrast, the tenets of religion are truly based on faith, since there is no empirical data to support them.

2. Davies claims that the "faith" of science is based on something outside the universe, like "an unexplained set of physical laws. " The lack of a current explanation for why the laws are as they are, however, does not make physics a faith. It only means that we don't have the answer. Indeed, Davies thinks we might be able to come up with an answer, one that does not involve supernatural intervention. So what, exactly, are scientists taking on faith here? What do we believe to be true without any evidence? I don't get it.

Nathan Myhrvold

Response to Paul Davies

Paul Davies has done some interesting scientific work, and written some engaging popular books. However I think his NYT op ed is way off target. It is a brave attempt to carve out some intellectual respectability for religious beliefs.

Like many bad arguments, it is based on some kernels of truth, and that is worth examining. Small elements of faith do crop into science generally, and physics in particular.

A priori, there is no reason to believe that the universe has simple laws of physics. The entire endeavor of physics is based on the belief—Paul would say faith—that:

(a) The universe is governed by a set of laws. (b) We humans can figure them out.

On an individual basis, each physicist also has a third form of faith:

(c) That through my own hunches, guesses and hard work, I can figure out some aspect of physical law.

Every discovery or invention is an unproven hunch or guess. You have to have faith in your own abilities, and your work, to move forward. But that is not capital "F" Faith of religion—it is the pragmatic working assumption that it is worth believing in yourself. Frankly, proposition (c) is far more important to a working scientist than (a) or (b). Any rational scientist with a bounded ego has to conclude that he or she is likely to be weak link in the chain. There may be a physical law, but are you going to be the one to understand it? Or, to be more mundane, will your funding grant be approved so you can even ask?

I think Paul is correct when he says that physicists have faith that (a)–(c) are true. It didn't have to be this way. The great physicist and Nobel Prize winner Eugene Wigner wrote a brilliant essay on this many years ago entitled "The Unreasonable Effectiveness of Mathematics in the Natural Sciences".

So I will concede that every physicist has (implicitly at least) faith that these propositions hold, else why waste your time doing physics? However, Paul conveniently overlooks the fact that time and time again, both propositions have indeed proven to be true! Furthermore, they are put to the test repeatedly by a process in which they can fail. A working physicist assumes that (a) and (b) hold because there is a 500 year history of propositions and tests. The net result is what we know about physics, which describes the world very well indeed.

The entire methodology of physics is to repeatedly test those propositions—both in the general sense, and specifically in terms of many individual theories put forth by

physicists. If nobody could make sense of the laws of the universe, physicists would have given up. If mathematics had not proven to be an effective language for describing the world (in Wigner's sense) then people would have eventually lost faith and given up. As indeed they have given up on thousands of theories and hypotheses over the years which failed the acid test of experimental or observational verification.

To argue that physics is on the same level as religion is to ignore the fact that religious faith broadly does not admit that it could be wrong! Consequently, it performs no experiments or observations. Religion is a broad topic of course, and there are myriad disparate theologies and philosophies so it is impossible to generalize too much. However, I am not aware of any religion that repeatedly tests its propositions—either in terms of small details, or the broad foundations—with experiments and observations. Surely this is not the case with any Christian theology that I am aware of.

There is a small and instructive exception in the case of Catholic saints. Non-martyrs require "proof" of one "miracle" to reach beatification. A second miracle is required to move up from beatification to full canonization. I find it quite interesting that empirical observational evidence is part of a religious process. In principle at least, it is an evidence based system. Both beatifications and canonizations have been refused on this basis (at least outwardly).

Yet the role of observational evidence in this case is the proverbial exception that proves the rule. The rules of evidence for miracles are essentially non-existent. Non-miraculous explanations are not allowed to compete. The last thing that the church would allow would be a rigorous double-blind test of saintly efficacy. Once you started down that path, some doubting Thomas might ask about all of the other religious propositions. Why should any of religion be believed without proof?

The modern theological answer is that Faith—believing something with no good reason—is somehow essential as a bedrock formation of religion. It is a bold strategy to take this weakness and promote it as a strength. It is also a modern affectation—efficacy was the goal in most early religions. People prayed and sacrificed to their gods for all sort of very practical real world purposes because they thought it would work. Over time theologians have retreated from the practical to spiritual domains, staying one step ahead of the law, so to speak.

At this stage the faith required by a physicist is not whether propositions (a) and (b) are true, but rather will they remain to be true in the future. They are demonstrably true for every part of physical law that we have figured out to date. In this case, "figured out" means "the theory matches experiment or observation very well".

To be fair, the past successes do not mean that they continue in the future. Perhaps it is only the simple stuff (or to be more precise, the low energy limit or some such) where it seems like physical laws exist. Or perhaps they exist but we're too dumb to figure them out, or can't get enough funding for the experiments. John Horgan wrote a book called *The End of Science* a few years ago arguing that modern particle physics may be in this

situation. I disagree with Horgan, but it certainly is possible. Indeed my disagreement is a statement of faith that by continuing to investigate we'll push forward.

Cosmology and fundamental particle physics are the branches of physics where these questions tend to be asked. It is pretty damn hard for a solid state physicist to get philosophical about what is happening inside a semiconductor—even though the same principles could crop up. Paul is a cosmologist, and so was I when I did physics. Cosmology asks questions like what is the fundamental nature of space and time? Where did the universe come from? These questions carry with them an unfortunate baggage that distract people with navel gazing and worrying about philosophical issues. Time and time again, even Cosmology has proven that the scientific method works—we know vastly more about the early Universe now than in the past. The baggage is annoying (and sometimes fun to speculate about) but ultimately science has triumphed.

Paul's overall approach is to try to get the camel's nose inside the tent by saying "science is not without faith". I say so what? It is ridiculous to the repeatedly tested faith of science represented by (a) through (c) at the same level of the blind faith as religion, which has yet to face even a single good experimental test. Instead it has a 2000 year history (in the case of Christianity) history of failing to predict experiments. Why does this even matter? Indeed that is my final rejoinder to Paul. Yes, there is a certain measured, bounded, and repeatedly tested faith in science. That stands in sharp contrast to the unbounded and untested faith in religion. The two are not the same—indeed the way each field treats that faith is the telling point.

Lawrence Krauss

Response to Paul Davies

Einstein once said that what most interested him about the Universe was whether God had any choice in its creation. He was, of course, not referring to a deity here, but rather asking the very important question: Can there only be one set of physical laws that allow for a consistent physical universe, or are there many possibilities?

This is precisely the question that Paul Davies suggests scientists do not generally ask, and moreover it demonstrates profound differences between the 'faith' of scientists, and religious faith. It is true that there is no purpose in carrying out scientific investigations if we are to believe that the laws of nature are capricious and can change from day to day in unpredictable ways. But this kind of faith is like having faith that because the sun rose at 6:57 this morning, it will rise at a predictably close time tomorrow. It is almost an insult to religious faith to suggest that this faith is on the same plane as faith in a divine presence who has endowed the universe with purpose and design, and in the case of most modern religions, is also vitally concerned with the day to day tragedies of humanity. It also misrepresents the scientific process.

The faith in something outside the universe, described by Davies as a common property of science and religion plays a central role in religion—God is the center of existence—whereas to the extent that scientists accept the existence of physical laws as being given, it is essentially peripheral to the everyday workings of science.

Moreover, the facts that (a) the scientific method continually refines and changes our understanding of physical law, whereas religious 'truths' have remained largely unchanged, and (b) scientists are now, as Davies mentions, trying to address questions of the origin of physical law, both suggest the comparison that Davies is trying to make between science and religious faith is strained at best.

Scott Atran

Response to Paul Davies

The scientific revolution began in earnest when a Polish cleric, Nicolaus Copernicus, bucked his faith and theorized that the earth turned around the sun. The Church did not pay much mind as long as the theory remained in the realm of speculation. But when Italian philosopher Galileo Galilei empirically confirmed the theory with a telescope, the Church banned Copernicus's teachings as "false and altogether opposed to the Holy Scripture. " In 1633, Galileo himself was brought to trial by the Holy Inquisition and compelled to recant.

Given the supposed risk of society's moral degradation in the face of the free choice to make up one's own mind ("I think, therefore I am"), the Church violently insisted that ideological faith in absolute authority ("In the beginning, God created the heavens and the earth") must always trump the more tentative teaching that goes with clear reasoning and experimental observation. When the Enlightenment unshackled scientific thinking from lingering religious control, religion opted for a separate realm where science would not operate. Science, for the sake of its peace and independence, generally accepted this division into separate "Magesteria. " In 1992 the Catholic Church cleared Galileo's name and in 2000 Pope Jean Paul II apologized to God (not to Galileo) for the trial.

Today, however, increasingly many on the science side argue that this separation was at best a temporary armistice to gather force without hindrance from religion, at worse an act of cowardice and partial capitulation. In any event, science should now invade the religious realm in order to conquer it and make people less superstitious, more knowledgeable, and happier. Davies agrees that the division of religion and science into separate Magesteria is wrong, not because science should be allowed to triumph but because scientists must squarely face the fear that their triumphalism will be undermined by the acknowledgment that they are as bound by faith as religious believers.

While it is true that many scientists are inductively-driven (rather than authority-driven) by faith in Nature's ultimate regularity, which the experimental mode of knowing tends to repeatedly confirm, others do not rely on such faith in the regularity of the universe to justify science. Immanuel Kant, for example, thought that science was the response to structures imposed on experience by human minds. We see of the world what the mind allows us to see, regardless of how the world really is. Science is successful as long as we can continue to navigate the world with the regularity our minds impose. He did not think we could, or need, to know anything about the ultimate character of reality to do good and effective science. For all Kant cared, the intrinsic organization of the world could be anarchic.

Paul Davies, argument that science is based on faith is not new. The chemist and philosopher Michael Polanyi argued that scientific discovery is only possible because of a scientist's faith that an unknown discovery is possible. The scientist's willingness to commit significant resources and time to achieving a result that is unknown at the outset, and before the benefits of discovery are imagined, depends on the non-demonstrable belief that basic laws pattern the universe and that humans are freely and uniquely (at least on this planet) capable of noticing and investigating evidence of this patterning.

But scientific faith in universal law is not dogmatic. While a scientist must make presuppositions in order to get started, everything is revisable and discardable, even belief in the regular patterning of the universe. Einstein, like Newton before him, believed that the universe was structured with deterministic mathematical regularity; Bohr and most of Einstein's later colleagues did not. Einstein did not ignore Bohr, or want to try him or burn him at the stake, but continued to argue with him and to provisionally accept his findings, even as Einstein struggled on in the faith that there are all-encompassing laws that mathematically determine all events in the universe.

Physicist Murray Gell-Mann, like most modern physicists, believes Einstein was off track in his later life. But on the larger issue of whether or not the universe is built according to basic mathematical laws (plus a lot of accidents) the jury is still out. Perhaps, as Newton said and Gell-Mann also believes, "Nature is very consonant and conformable to herself." The upshot of this for science is that it succeeds in getting closer and closer to Nature's underlying structure because, like the successive layers of an onion, each of Nature's layers that is mathematically peeled away by science closely conforms to the next underlying layer and mathematical theory. Evidence for this is the progressive elegance and simplification of mathematical theories and the simultaneous broadening and deepening of their empirical scope (e.g. Lorentz to Einstein, with Einstein arguing that he was driven to his theory of relativity by the lack of elegance in previous theories). The hypothesis is that with scientific discipline and mathematics, human minds are able to progressively capture Nature's onion-like structure because that structure, which includes the structure of the mind itself, is basically mathematically regular (with lots of accidents).

There is an evolutionary just-so story (perhaps true) that may help to make sense of this hypothesis. Humans are able to count because the world contains countable things (like stars, stones, and seasons) that can be put in successive correspondence with one another, and because the mind evolved to capture what the universe contains so that the organism that carried this representational tool (the mind) could survive and reproduce. The evolution of language—of discrete relationships between sounds and meanings that refer to things in the world—likely provided the infinite discreteness required for formulating any mathematical system. Since human minds can grasp indefinitely many mathematical systems, the difficulty is in figuring out precisely which mathematical structure adequately represents which aspect of Nature. Experiments and empirical intuition help to weed out and whittle down the candidate mathematics.

One empirical prediction for the plausibility of this hypothesis is that universal patterns of physical and biological regularity will have likely produced many other forms of intelligent life in the cosmos. All would presumably be "conformable with Nature" so that any mathematical description of physical phenomena used by an alien technological civilization on another planet would likely resemble what human beings come up with on Earth, however different the notation. The search for confirmation of this prediction would be guided by a faith that sores beyond religion, but not beyond reason.

Sean Carroll

Response to Paul Davies

Why do the laws of physics take the form they do? It sounds like a reasonable question, if you don't think about it very hard. After all, we ask similar-sounding questions all the time. Why is the sky blue? Why won't my car start? Why won't Cindy answer my emails?

And these questions have sensible answers—the sky is blue because short wavelengths are Rayleigh-scattered by the atmosphere, your car won't start because the battery is dead, and Cindy won't answer your emails because she told you a dozen times already that it's over but you just won't listen. So, at first glance, it seems plausible that there could be a similar answer to the question of why the laws of physics take the form they do.

But there isn't. At least, there isn't any as far as we know, and there's certainly no reason why there must be. The more mundane "why" questions make sense because they refer to objects and processes that are embedded in larger systems of cause and effect. The atmosphere is made of atoms, light is made of photons, and they obey the rules of atomic physics. The battery of the car provides electricity, which the engine needs to start. You and Cindy relate to each other within a structure of social interactions. In every case, our questions are being asked in the context of an explanatory framework in which it's perfectly clear what form a sensible answer might take.

The universe (in the sense of "the entire natural world," not only the physical region observable to us) isn't like that. It's not embedded in a bigger structure; it's all there is. We are lulled into asking "why" questions about the universe by sloppily extending the way we think about local phenomena to the whole shebang. What kind of answers could we possibly be expecting?

I can think of a few possibilities. One is logical necessity: the laws of physics take the form they do because no other form is possible. But that can't be right; it's easy to think of other possible forms. The universe could be a gas of hard spheres interacting under the rules of Newtonian mechanics, or it could be a cellular automaton, or it could be a single point. Another possibility is external influence: the universe is not all there is, but instead is the product of some higher (supernatural?) power. That is a conceivable answer, but not a very good one, as there is neither evidence for such a power nor any need to invoke it.

The final possibility, which seems to be the right one, is: that's just how things are. There is a chain of explanations concerning things that happen in the universe, which ultimately reaches to the fundamental laws of nature and stops. This is a simple hypothesis that fits all the data; until it stops being consistent with what we know about the universe, the burden of proof is on any alternative idea for why the laws take the form they do.

But there is a deep-seated human urge to think otherwise. We want to believe that the universe has a purpose, just as we want to believe that our next lottery ticket will hit. Ever since ancient philosophers contemplated the cosmos, humans have sought teleological explanations for the apparently random activities all around them. There is a strong temptation to approach the universe with a demand that it make sense of itself and of our lives, rather than simply accepting it for what it is.

Part of the job of being a good scientist is to overcome that temptation. "The idea that the laws exist reasonlessly is deeply anti-rational" is a deeply anti-rational statement. The laws exist however they exist, and it's our job to figure that out, not to insist ahead of time that nature's innermost workings conform to our predilections, or provide us with succor in the face of an unfeeling cosmos.

Paul Davies argues that "the laws should have an explanation from within the universe," but admits that "the specifics of that explanation are a matter for future research." This is reminiscent of Wolfgang Pauli's postcard to George Gamow, featuring an empty rectangle: "This is to show I can paint like Titian. Only technical details are missing." The reason why it's hard to find an explanation for the laws of physics within the universe is that the concept makes no sense. If we were to understand the ultimate laws of nature, that particular ambitious intellectual project would be finished, and we could move on to other things. It might be amusing to contemplate how things would be different with another set of laws, but at the end of the day the laws are what they are.

Human beings have a natural tendency to look for meaning and purpose out there in the universe, but we shouldn't elevate that tendency to a cosmic principle. Meaning and purpose are created by us, not lurking somewhere within the ultimate architecture of reality. And that's okay. I'm happy to take the universe just as we find it; it's the only one we have.

Jeremy Bernstein

Response to Paul Davies

Paul Davies's donnish question which he apparently tries out on the odd scientist—explain the laws of the universe—seems silly to me. Explain in terms of what?

PZ Myers

Response to Paul Davies

I'm not sure why the New York Times saw fit to publish Paul Davies's curious op-ed, except that Davies does have a reputation as a popularizer of physics, and as something of an apologist for deism; they certainly couldn't have chosen to print it on its merits. His argument is the tired and familiar claim that science has to be taken on faith, so it's just like religion. I recall hearing variants on this back in the schoolyard, usually punctuated with "nyaa nyaas" and assertions about each others' mothers, and while we may not have said much about science, the principle was the same. Citing a false equivalency is a cheap argument, but not very credible.

Davies lost my respect for his thesis early on, from the first sentence actually, but I'll focus instead on this claim from his second paragraph:

All science proceeds on the assumption that nature is ordered in a rational and intelligible way. You couldn't be a scientist if you thought the universe was a meaningless jumble of odds and ends haphazardly juxtaposed.

Perhaps this is where not being a physicist has the virtue of a different perspective, because I can say without reservation that he's completely wrong—in a historical science like evolutionary biology, we have no problem when we encounter a phenomenon that isn't orderly or rational, and that has all the appearance of haphazard meaninglessness. We're accustomed to seeing simple chance as a strong thread running throughout biological history.

Pattern and order are important too, of course, but when looking at the appearance of some particular feature we have to be prepared for the possibility that it is not a consequence of some orderly progression—perhaps it just happened that way. I can't imagine that my physicist colleagues are any different, and that they would be horrified to discover that physical order was "rooted in reasonless absurdity". That would be interesting. If that is the way the universe is, that is what science will try to grapple with (admittedly, we might have serious difficulties grappling with total chaos, but no one claims that science can have answers for everything). That Davies seems to believe that order must rule everywhere and at every level is a stronger presupposition than is warranted by a scientific approach, and sounds remarkably theological...and I don't think Davies would object to the charge of theology, although he clearly thinks the only good science fits his theological model.

But then Davies does have this notion that that the concept of physical laws is derived from Christian doctrine—that science is rooted in attempts to define the actions of a supernatural lawgiver who imposes a kind of universal consistency on everything. As a

historical argument, and as a psychological description of the way the minds of people like Newton worked, I can go along with that; but as an assumption that this expectation of a universal order must reflect a universal reality, I disagree. If the laws of physics were subtly different in Egypt than in Greece, we would have developed an empirical physics that took that into account; that certain laws are constant everywhere is just what is, as empirically determined by scientific observation. A geologist, a biologist, an anthropologist, and a historian will also be able to tell you that there are many things that are quite different between Egypt and Greece, and yet variation does not mean those sciences fail.

Alas, Davies also brings up the anthropic principle, that tiresome exercise in metaphysical masturbation that always flounders somewhere in the repellent ditch between narcissism and solipsism. When someone says that life would not exist if the laws of physics were just a little bit different, I have to wonder... how do they know? Just as there are many different combinations of amino acids that can make any particular enzyme, why can't there be many different combinations of physical laws that can yield life? Do the experiment of testing different universes, then come talk to me. Until then, claiming that the anthropic principle, an undefined mish-mash of untested assumptions, supports your personal interpretation of how the universe exists and came to be is a self-delusional error.

I'm also always a bit disappointed with the statements of anthropic principle proponents for another reason. If these are the best and only laws that can give rise to intelligent life in the universe, why do they do such a lousy job of it? Life is found in one thin and delicate film on one planet in this mostly empty region of space, and even if there are other fertile planets out there, they will be nearly impossibly distant, and life will be just as fragile and prone to extinction there as here. Even on this world, all of the available environments favor bacteria over scientists or theologians, and said scientists and theologians have existed for only about 0.00001% of the lifetime of this universe, and are prone to wink out of existence long before we can get rid of one of the zeroes in that number.

If I wanted to argue for a position on the basis of the anthropic principle, rather than trying to pretend that we live in a Goldilocks universe, we should be wondering how we ended up in such a hostile dump of a universe, one that favors endless expanses of frigid nothingness with scattered hydrogen molecules over one that has trillions of square light years of temperate lakefront property with good fishing, soft breezes, and free wireless networking.

Maybe Davies has faith in science, but I don't. I take it as it comes. I have expectations and hypotheses, but these are lesser presuppositions than what is implied by faith—and I'm also open to the possibility that any predictions I might make will fail. Perhaps if Davies weren't so obsessed with equating his religion with his science, he wouldn't be blind to the fact that most scientists don't see his god in the operation of the universe.

Lee Smolin

Response to Paul Davies

Davies' essay has engenders a confused response because it seems to be making two different arguments. In the first part he seems to be advocating equating science with religion. It is hard to know how else to read his assertion that "Clearly, then, both religion and science are founded on faith—namely, on belief in the existence of something outside the universe..." Certainly one has to oppose this for all the reasons the other respondents have pointed out.

What becomes clear only towards the end of his essay is that he is advocating a way for science to avoid the equation with religion. This is to give up the belief in eternal, immutable laws in favor of a notion that "the laws should have an explanation from within the universe and not involve appealing to an external agency." This is opposed to the notion of eternal immutable laws that Davies notes has common roots and logic with the Christian view of God.

To understand the problems that Davies is addressing-and to understand why they do not concern most of science—one has to appreciate that the questions he is dealing with arise only when we attempt to ask scientific questions about the universe as a whole. Only then do we have to address two key questions: "Why these laws?" and "Why these initial conditions for the universe?"

Davies point is that if we just accept the laws and initial conditions of the whole universe as just given, with no further explanation, then we are reducing science to faith in the unexplained. He is right about this, but he doesn't emphasize that these questions are avoided in most domains of science, where we study isolated systems in a larger universe. In these cases we can cleanly separate regularities into laws and initial conditions. This gives an operational meaning to the notion of fixed laws. It is only when we extend our reach to the universe as a whole, that the distinction between laws and initial conditions becomes problematic. For example, there are observations, such as wiggles in the CMB spectrum, that can be explained either by modifying laws or modifying initial conditions and, since there is only one instance of the universe, there is no way to determine which is correct.

This suggests, however, the need for a better notion of law, applicable on a cosmological scale, not for a surrender to religion or metaphysics.

Indeed, there are reasons to worry that contemporary cosmological theory has wandered from the domain of science. Reliance on infinite and eternal ensembles of universes, and use of the anthropic principle, popular as they have been in the recent literature, have not led to a single falsifiable prediction. There is a real danger of

unjustified belief in putative unified theories that cannot be experimentally tested.

But these are not the only possible responses to the questions of "Why these laws?" and "Why these initial conditions?". They are only the result of approaches to them that fail to critically ask what notion of law is appropriate for doing science on a cosmological scale. In fact there are approaches to these questions that, if they succeed, will bring progress to science, in the conventional sense that it will lead to new predictions which are falsifiable or verifiable by doable experiments

There is a healthy skepticism about the role of philosophy in science, but when it comes to this issue of "Why these laws?" there is wisdom to be gained from studying philosophers who anticipated that science would reach the point of asking this question. Leibniz postulated two principles to address it, which are his "Principle of sufficient reason" the "Principle of the identity of the indiscernible." The first posits that we live in a universe in which every question of the form of "Why these laws?" or "Why these initial conditions?" has a rational answer. The second implies that there are no symmetries in the fundamental laws of nature. These principles are the underpinnings of the relational approach to space and time which culminated in general relativity.

Indeed, before relativity there was believed to be a law of nature of the form, "The geometry of space is three dimensional Euclidean geometry." People, for example, Kant, try to explain, "Why this law?" But in fact there was a perfectly rational answer, on which much further progress has been based. In general relativity the geometry of space is not determined by a law, it is a contingent and changing fact. The geometry at present can be understood to be the result of lawful evolution from past conditions. This is an example of how a deeper law can be discovered which moves an apparently eternal law to the status of contingent and fully explainable. This shows that the idea of a universe determining its own properties by dynamical evolution is not senseless, as Carroll suggests. It has been fully realized in general relativity and could be so again.

The American pragmatist Charles Sanders Pierce addressed this issue when he wrote in 1893, "To suppose universal laws of nature capable of being apprehended by the mind and yet having no reason for their special forms, but standing inexplicable and irrational, is hardly a justifiable position. Uniformities are precisely the sort of facts that need to be accounted for. Law is par excellence the thing that wants a reason." He went on to assert that, the only possible way of accounting for the laws of nature, and for uniformity in general, is to suppose them results of evolution ."

Might this apply to contemporary physics and cosmology? In 1992 I proposed a cosmological scenario based on natural selection that explained a great deal about "Why these laws" while leading to several falsifiable predictions for real astrophysical observations. It was based on the then novel idea of a landscape of theories whose parameters evolve by a mechanism formally similar to population biology. In the 15 years since, observations relevant for those predictions continue to be done and they continue to remain unfalsified. Meanwhile since 2003 the study of the landscape has become a big

fashion, but most choose to study it using a combination of timeless probability distributions and the anthropic principle. This, as I have argued in detail elsewhere, cannot lead to any falsifiable predictions and indeed, despite a literature of hundreds of papers, it has not. Pierce seems more and more to have been vindicated.

I do not know if the particular cosmological scenario I proposed is correct, but since it is a real scientific theory, vulnerable to falsification, we will know sooner or later. What its existence does demonstrate is that there is no need for big pronouncements that equate scientific methodology with religious faith. Science can proceed, even when faced with the questions of "Why these laws" and "Why these initial conditions". We just have to adopt a strategy that leads to predictions testable by real experiments.

Biologists used to ask, "Why these species?" They found they could answer it if they gave up the idea that species are immutable and accepting the idea that species and their characteristics are contingent outcomes of the dynamical process of natural selection. In physics the evidence points to the same conclusion. This means that regularities we used to think of as immutable laws, have to be thought of as subject to processes of evolution. This makes better science because, like biology, it makes hypotheses as to the mechanism of evolution subject to testing by observations.

There is, among theorists of cosmology and physics, some resistance to giving up the notion of immutable laws. This is irrational because the cost, as the last five years has made increasingly clear, is an embrace of strategies to rescue immutable laws such as the anthropic principle and eternal inflation, which do not lead to any falsifiable predictions. (Claims to the contrary can be easily shown to be fallacious.) As Davies says, the idea of immutable eternal laws arose in Newton's time, when science and theology were much closer together. It does seem that getting rid of this idea is a necessary step for modern science to become fully free of the 17th Century theological climate in which it was born.

However it is an exaggeration to say that until this is done science's claims to be "free of faith are bogus." Most of science is healthy and most scientists rely on notions of laws that are restricted to certain domains where they are well tested. It is only in fundamental theoretical physics and cosmology that we face the challenge of inventing new kinds of laws and explanations that do not carry the baggage of the theological roots of physics.

John Horgan

Response to Paul Davies

Paul Davies is a nice guy and graceful writer, but this essay reads like a parody of *The Mind of God* and other writings in which he glimpses fuzzy convergences between science and religion. Davies has sold lots of books peddling this theme, and he has won the \$1,000,000-plus Templeton Award, created by the financier John Templeton to help bridge the divide between science and religion. I mention the Templeton moola because, as I've said previously, I believe it is distorting all this science-religion "dialogue."

In his New York Times OpEd, Davies notes that "both monotheistic religion and orthodox science fail to provide a complete account of physical existence." While scientists have done a good job discovering basic laws of nature, they can't explain where the laws came from in the first place. They have tentative ideas—for example, involving other universes with vague connections to ours—but nothing supported by evidence, Davies says.

I've knocked multiverse theories too, and in fact I suspect we'll never know why our universe is the way it is. Many scientists, however, are optimistic that they will answer this question eventually. Noting that fact, Davies asserts that science and religion both "are founded on faith—namely, on belief in the existence of something outside the universe, like an unexplained God or an unexplained set of physical laws, maybe even a huge ensemble of unseen universes, too." Unless science can serve up a testable theory of where the universe and its laws came from, Davies concludes, science's "claim to be free of faith is manifestly bogus."

Huh? Science works. Scientists have gleaned deep insights into nature through experiment, reason and hard work. These tangible, world-transforming achievements inspire scientists' optimism—or "faith," to use Davies's term—that they will go even further. Religion has not told us one demonstrably true thing about nature. Not one! Religion is nothing BUT faith.

The only thing that's bogus here is Davies's analogy between science and religion.

Alan Sokal

Response to Paul Davies

Here is my letter to the editor to the New York Times about Davies' op-ed, which they chose not to print, perhaps because it was too blunt for their taste:

To the editor:

Paul Davies' claim (op-ed, Nov. 24) that "both religion and science are founded on faith" is based on astoundingly sloppy reasoning.

Science is, indeed, founded on the working hypothesis—one amply borne out by four centuries of scientific practice—that the world, or at least some aspects of it, is ordered in a stable and intelligible way. But that tentative and partly testable working hypothesis is a far cry from religions' reliance on sacred texts and personal revelations. To characterize these radically dissimilar endeavors as both based on "faith" is to point out a superficial commonality while obscuring the fundamental difference. And at a time when humanity is wracked by conflict between incompatible versions of faith—in the genuine sense of the term—to muddy the distinction

between religion and science is worse than philosophically misguided: it is irresponsible.

Paul Davies (Response)

The author replies to his critics

I was dismayed at how many of my detractors completely misunderstood what I had written. Indeed, their responses bore the hallmarks of a superficial knee-jerk reaction to the sight of the words "science" and "faith" juxtaposed.

The most common trap my critics have fallen into is in conflating the explanation of natural phenomena using the laws of physics, with an explanation of the laws themselves. I am not suggesting that the application of science is a matter of faith. Doing science involves employing testable hypotheses, refining theories and conducting experiments—in stark contrast to the practice of religion. The scientific method is the most reliable path to truth we know, and there is no more committed or passionate a scientist than I. Yes, "science works" as John Horgan points out. It is tested again and again as a description of nature. We are all agreed on that point. But it isn't the point I was trying to make. My argument refers, not to the scientific method, but to the underlying lawfulness of the universe itself, which raises questions such as where the laws come from, why they have the form that they do, and whether there is anything peculiar about the actual laws of the universe (such as being "fine-tuned" for life), as opposed to other possible laws. The orthodox position (and the one I set out to challenge in my book) is that the universe is governed by a fixed set of laws in the form of infinitely precise mathematical relationships imprinted on the universe from its birth. In addition, it is assumed that the physical world is affected by the laws, but the laws are completely impervious to what happens in the universe—they are immutable. It is not hard to see where this picture comes from: it is inherited from monotheism, which asserts that a rational being designed the universe according to a set of perfect laws. And the asymmetry between immutable laws and contingent states mirrors the asymmetry between God and nature: the universe depends utterly on God for its existence whereas God's existence does not depend on the universe. Historians of science are well aware that Newton and his contemporaries believed that in doing science they were uncovering the divine plan for the universe in the form of its underlying mathematical order. I am depressed that reminding scientists of this well-known historical fact should elicit such a shock-horror response. As Scott Atran points out, the argument that science is based on faith is not new. Evidently Western society is so steeped in monotheism that the monotheistic world view, which was appropriated by science, is now regarded as "obvious" and "natural." As a result, many scientists are unaware of its theological origin. Nor do they stop to think about the sweeping hidden assumptions they adopt when they subscribe to that scientific/theological world view, assumptions that are in fact are not shared by most other cultures.

Not all scientists envisage the laws of nature in the theological manner I have described, however. One person who evidently doesn't is P.Z. Myers, who declares his a lack of faith in science and simply takes science "as it comes." I have found that his is a familiar position among biologists, for whom contingency as opposed to law looms so large in explanation. Unfortunately, Myers goes on to attribute to me precisely the point of view I am seeking to refute: "That Davies seems to believe that order must rule everywhere and at every level is a stronger presupposition than is warranted by a scientific approach, and sounds remarkably theological." Well, yes, that's the whole point of my article! It is theological—but it is nevertheless the orthodox view among theoretical physicists, especially those working on the search for a unified theory. Such physicists believe there are perfect laws "out there", existing in some Platonic realm, even if the laws we find in our textbooks today are merely approximations to what Steven Weinberg calls "the final theory". And that is the position that, contrary to Myers' statement, I seek to challenge in my book. In doing so, I encountered fierce opposition from my physics colleagues. For example, when I suggested in my book that infinitely precise mathematical laws might be an unjustified idealization, i.e. that there might be an intrinsic uncertainty or flexibility in the laws, many of my physics colleagues were aghast at this heresy. Jerry Coyne, in his response to my article, asks, "What do we [orthodox scientists] believe to be true without evidence?" Well, how about belief in infinitely precise laws which incorporate real numbers and differentiability? Show me the evidence for that. Or, to take another well-known example, laws that transcend the physical universe and exist in some sense prior to it, because the said laws are intended to explain the origin of the universe. Many cosmologists believe in such laws, which must be accepted without explanation or testability, as the basis of a scientific theory of cosmogenesis.

My article pointed out that the widespread belief in immutable perfect transcendent prior laws underpinning the physical universe, while not necessarily wrong, is nevertheless held as an act of faith, similar in character to belief in an all-perfect divine lawgiver. Let me be clear about the sense in which I am using the word faith here. Obviously faith in the laws of physics isn't on a par with "faith" in the popular religious sense (such as belief in miracles, prophecy, the bible as historical fact, etc., all of which I personally regard as completely ridiculous). Rather, in using the word faith I refer to the metaphysical framework, shared by monotheism and science (but not by many other cultures), of a rational ground that underpins physical existence. It is the shared faith that we live in a universe that is coherent, a universe that manifests a specific mathematical scheme of things, a universe that is, at least in part, intelligible to sentient mortals. These tacit assumptions running through science, that stem from monotheism, can all be challenged. The universe doesn't have to be that way! But most scientists believe it is that way.

Because the monotheistic world view pervading science is so deeply entrenched, asking where the laws of physics come from or why they have the form they do is frowned upon. Many respondents to my article ticked me off for venturing into such murky waters, or for expecting there to even exist an answer. I am grateful to Sean Carroll for so cogently

expressing the orthodox view among physicists that the laws of physics must simply be accepted as a brute fact—that is, they exist without explanation, for no reason. "That's just how things are," writes Carroll. "There is a chain of explanations concerning things that happen in the universe, which ultimately reaches to the fundamental laws of nature and stops." For Carroll, as for many scientists, unexplained laws are thus the starting point of scientific reasoning, the levitating superturtle at the bottom that holds up the whole tower, just as God is the levitating superturtle that holds up physical existence in monotheism. After 30 years of listening to sterile bickering in the science/religion debate I am utterly bored with the refrain from each side that, in effect, "my superturtle is better than your superturtle." So I have tried to elevate the level of discussion and move on.

To achieve progress, I set out to see how far we can go in describing the deepest properties of the physical universe without appealing to anything outside it—such as an unexplained transcendent god, an unexplained set of magically-imposed Platonic mathematical laws, or infinitely many unseen alternative universes. I concede that, so long as we are stuck with human modes of thought, we will ultimately have to accept something on faith, but I see no reason to stop with the laws of physics. So I question the idealized concept of immutable perfect laws that must simply be accepted as a brute fact—"on faith" (since we can test the laws only to finite precision). The time has come to seek a theory of the laws, to bring the laws of physics within the scope of scientific inquiry, and if possible to explain their intelligibility, their "unreasonable" mathematical efficacy and their celebrated (and baffling) bio-friendliness. A possible way to formulate a theory of laws was mooted thirty years ago by John Wheeler, who abandoned the traditional theological notion of immutable laws "cast in tablets of stone from everlasting to everlasting." In my book I have sought to extend Wheeler's ideas in the light of recent work in the foundations of quantum mechanics, the theory of computation and holographic cosmology. Another possible approach to a theory of laws has been developed by Lee Smolin, and is mentioned briefly in his response to my article.

My interest in pursuing this project is to critically examine ultimate explanations of existence, for which there is a long tradition within religion, and a rather short one within science. I plead guilty to Lawrence Krauss' complaint that I am sidestepping some hugely important issues, such as the moral dimension of religious faith, the tragedy of human existence and suffering, and the question of purpose in the universe. My concern is admittedly with a restricted physics/cosmology agenda, as that is the only area in which I can claim some modest authority. However, the conceptual framework I am developing can accommodate a universe with something like "purpose," albeit one that is inherent in, and emergent with, the universe, rather than imposed upon it from without.

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